Chapter 1: The Fundamental Unit of Life

1.1 Introduction to Cell Biology

Cell biology is the study of cells, their physiological properties, their structure, the organelles they contain, interactions with their environment, their life cycle, division, death and cell function. This is done both on a microscopic and molecular l-evel.

Knowing the components of cells and how cells work is fundamental to all biological s-ciences.

It is also essential for research in bio-medical fields such as cancer, and other dis-eases.

Research in cell biology is closely related to genetics, biochemistry, molecular biol-ogy,
immunology, and developmental biology.

1.1.1 The Cell Theory

The cell theory, first developed in 1839 by Schleiden and Schwann, states that:

1. All living organisms are composed of one or more cells.
2. The cell is the basic unit of structure and organization in organisms.
3. Cells arise from pre-existing cells.

1.2 Types of Cells

Cells can be subdivided into two major categories: prokaryotic and eukaryotic. The main difference between the two is that eukaryotic cells contain a nucleus and membrane-bound organelles, while prokaryotic cells do not.

1.2.1 Prokaryotic Cells

Prokaryotes include bacteria and archaea. They are much simpler than eukaryotes.
A prokaryotic cell has three architectural regions:
- Envelopes (capsule, cell wall, and plasma membrane).
- Cytoplasmic region (contains the genome or DNA).
- Appendages (flagella and pili).

1.2.2 Eukaryotic Cells

Plants, animals, fungi, slime moulds, protozoa, and algae are all eukaryotic. These cells are about fifteen times wider than a typical prokaryote and can be as much as a thousand times greater in volume. The main distinguishing feature is the presence of a membrane-bound nucleus.

1.3 Key Organelles

1.3.1 The Nucleus

The nucleus is the information center of the cell and houses the cell's chromosomes, which are the linear DNA strands. It is surrounded by a double membrane called the nuclear envelope.

1.3.2 Mitochondria

Mitochondria are often described as the "powerhouse of the cell" because they generat-
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most of the cell's supply of adenosine triphosphate (ATP), used as a source of chemic-
al energy.
In addition to supplying cellular energy, mitochondria are involved in other tasks, such as signaling, cellular differentiation, and cell death.

1.3.3 Ribosomes

The ribosome is a large complex of RNA and protein molecules. They act as an assembly-
line
where RNA from the nucleus is used to synthesize proteins from amino acids. Ribosomes-
can
be found either floating freely or bound to the endoplasmic reticulum.

1.4 Conclusion

Understanding cell structure and function is critical for modern biology.
From the simplicity of prokaryotes to the complexity of eukaryotic organelles like mitochondria and the nucleus, life is built upon these microscopic units.